

PROOF OF CIGLER'S CONJECTURE ON q -HOGGATT NUMBERS

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ABSTRACT. We prove the nonnegativity and palindromicity of a family of polynomials arising from q -Hoggatt numbers. The nonnegativity is derived from Stanley's (P, ω) -partition theory through a standard Young tableau formula, while the palindromicity is proved by an involution on rectangular standard Young tableaux. Our result confirms a conjecture of Cigler.

1. INTRODUCTION

A polynomial $p(x)$ is *palindromic* if the sequence of its coefficients reads the same forwards from the lowest power to the highest as it does backwards. In other words, $p(x)$ satisfies the condition

$$p(x) = x^N p(x^{-1}),$$

where N is the sum of the lowest and highest exponents. A classic example of palindromic polynomials is the q -binomial coefficient

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \begin{cases} \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}, & \text{if } 0 \leq k \leq n, \\ 0, & \text{otherwise,} \end{cases}$$

where the q -Pochhammer symbol is defined by

$$(a; q)_m := \prod_{j=0}^{m-1} (1 - aq^j).$$

As a polynomial in q , the q -binomial coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q$ has lowest exponent 0 and highest exponent $k(n-k)$, while it is a standard result [4, p. 353, eq. (I.47)] that

$$\begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{k(k-n)} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

In [2], Cigler introduced a q -analog of Hoggatt numbers originating from [3], and conjectured a palindromic property for a relevant series. Let

$$\langle n \rangle_q^{(d)} := \begin{bmatrix} n+d-1 \\ d \end{bmatrix}_q, \quad \langle n \rangle_q^{(d)}! := \prod_{j=1}^n \langle j \rangle_q^{(d)}.$$

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Cigler's q -Hoggatt numbers are defined by

$$\left\langle n \atop k \right\rangle_q^{(d)} := \frac{\langle n \rangle_q^{(d)}!}{\langle k \rangle_q^{(d)}! \langle n-k \rangle_q^{(d)}!} = \prod_{j=0}^{d-1} \frac{\left[\begin{smallmatrix} n+j \\ k \end{smallmatrix} \right]_q}{\left[\begin{smallmatrix} k+j \\ k \end{smallmatrix} \right]_q}. \quad (1.1)$$

For $d = 1, 2$, and 3 , the q -Hoggatt numbers $\langle n \rangle_q^{(d)}$ reduce to the q -binomial coefficients, the q -Narayana numbers, and the q -Baxter numbers, respectively.

Cigler [2, Conjecture 10] proposed the following conjecture on q -Hoggatt numbers.

Cigler's Conjecture. *Let d and k be any positive integers. The series*

$$C_k^{(d)}(x, q) := (x; q)_{kd+1} \sum_{n \geq 0} \left\langle n+k \atop k \right\rangle_q^{(d)} x^n \quad (1.2)$$

is a bivariate polynomial in x and q with nonnegative coefficients, with the highest exponent of x being $(k-1)(d-1)$. Writing

$$C_k^{(d)}(x, q) =: \sum_{j=0}^{(k-1)(d-1)} c_{k,j}^{(d)}(q) x^j,$$

it is further true that $c_{k,j}^{(d)}(q)$ is palindromic in q for every j .

The main purpose of this note is to prove Cigler's conjecture by means of Stanley's (P, ω) -partition theory [5–7].

Recall that an *integer partition* λ is a weakly decreasing sequence of positive integers known as *parts*; the size of this partition, denoted by $|\lambda|$, is the sum of all its parts. We may visually represent a partition $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_l)$ by its *Young diagram*, which is depicted by boxes placed in rows such that there are λ_i boxes in the i -th row.

Let λ be a fixed partition. A *standard Young tableau of shape λ* is a filling T of the boxes in the Young diagram of λ with $1, 2, \dots, |\lambda|$, each used exactly once, such that the entries are strictly increasing along rows and down columns. Let $\text{SYT}(\lambda)$ denote the set of such tableaux.

For $T \in \text{SYT}(\lambda)$ and $1 \leq i \leq |\lambda|$, let

$$\rho_i = (r_i, s_i) := T^{-1}(i)$$

denote the box occupied by i in T . Define the *descent set*

$$\text{Des}(T) := \{i : 1 \leq i \leq |\lambda| - 1, r_{i+1} > r_i\}.$$

We require the *descent statistic*

$$\text{des}(T) := |\text{Des}(T)|,$$

and the *comajor index statistic*

$$\text{comaj}(T) := \sum_{i \in \text{Des}(T)} (|\lambda| - i).$$

Our main result is to express Cigler's series $C_k^{(d)}(x, q)$ as the generating function for certain standard Young tableaux of rectangular shapes, thereby indicating

Cigler's conjecture. To start with, we let

$$(k^d) := \underbrace{(k, k, \dots, k)}_{d \text{ parts}},$$

the partition into d parts of size k . Its Young diagram has a rectangular shape with d rows and k columns.

Theorem 1.1. *For every positive integers d and k ,*

$$(x; q)_{kd+1} \sum_{n \geq 0} \left\langle \begin{matrix} n+k \\ k \end{matrix} \right\rangle_q^{(d)} x^n = \sum_{T \in \text{SYT}(k^d)} q^{\text{comaj}(T) - k \binom{d}{2}} x^{\text{des}(T) - (d-1)}. \quad (1.3)$$

Moreover, Cigler's conjecture is true. In particular, the coefficients at the lowest and highest powers of x are respectively

$$c_{k,0}^{(d)}(q) = 1, \quad c_{k,(k-1)(d-1)}^{(d)}(q) = q^{2 \binom{k}{2} \binom{d}{2}}.$$

This note is organized as follows. In Section 2, we interpret Cigler's series $C_k^{(d)}(x, q)$ combinatorially by proving (1.3); we also determine the coefficients of the two endpoint powers of x . Next, in Section 3, we establish the palindromicity for all coefficients $c_{k,j}^{(d)}(q)$; our argument is built upon a natural involution on rectangular standard Young tableaux. Finally, we close this paper with further discussions on Fibonacci–Hoggatt numbers and resolve another conjecture of Cigler in Section 4.

2. MAIN THEOREM, PART I: COMBINATORICS

In this part, we prove (1.3). Our starting point is the following observation.

Lemma 2.1. *For every nonnegative integer n and every positive integers d and k ,*

$$\left\langle \begin{matrix} n+k \\ k \end{matrix} \right\rangle_q^{(d)} = \sum_{T \in \text{SYT}(k^d)} q^{\text{comaj}(T) - k \binom{d}{2}} \left[\begin{matrix} (n+d-1) - \text{des}(T) + kd \\ kd \end{matrix} \right]_q. \quad (2.1)$$

Proof. We begin with a generic partition λ and a generic nonnegative integer m . Note that

$$\begin{aligned} & \sum_{T \in \text{SYT}(\lambda)} q^{\text{comaj}(T)} \left[\begin{matrix} m - \text{des}(T) + |\lambda| \\ |\lambda| \end{matrix} \right]_q \\ &= \sum_{T \in \text{SYT}(\lambda)} q^{\text{des}(T) \cdot |\lambda| - \text{maj}(T)} q^{(m - \text{des}(T)) \cdot |\lambda|} \left[\begin{matrix} m - \text{des}(T) + |\lambda| \\ |\lambda| \end{matrix} \right]_{q^{-1}}, \end{aligned}$$

where the *major index* of T is given by

$$\text{maj}(T) := \sum_{i \in \text{Des}(T)} i.$$

Thus,

$$\begin{aligned} & \sum_{T \in \text{SYT}(\lambda)} q^{\text{comaj}(T)} \left[\begin{matrix} m - \text{des}(T) + |\lambda| \\ |\lambda| \end{matrix} \right]_q \\ &= q^{m|\lambda|} \sum_{T \in \text{SYT}(\lambda)} q^{-\text{maj}(T)} \left[\begin{matrix} m - \text{des}(T) + |\lambda| \\ |\lambda| \end{matrix} \right]_{q^{-1}}. \end{aligned}$$

In light of [8, p. 364, Proposition 7.19.12], the sum on the right-hand side of the above can be expressed in terms of a *Schur function*:

$$\sum_{T \in \text{SYT}(\lambda)} q^{\text{comaj}(T)} \begin{bmatrix} m - \text{des}(T) + |\lambda| \\ |\lambda| \end{bmatrix}_q = q^{m|\lambda|} s_\lambda(1, q^{-1}, \dots, q^{-m}). \quad (2.2)$$

According to the above discussions, we have

$$\text{RHS (2.1)} = q^{kd(n+d-1)-k\binom{d}{2}} s_{(k^d)}(1, q^{-1}, \dots, q^{-(n+d-1)}).$$

Now we recall the principal evaluation of Schur functions [8, p. 374, Theorem 7.21.2] so that

$$\begin{aligned} s_{(k^d)}(1, q^{-1}, \dots, q^{-(n+d-1)}) &= q^{-k\binom{d}{2}} \prod_{i=1}^d \prod_{j=1}^k \frac{1 - q^{-(n+d+j-i)}}{1 - q^{-(k+d-i-j+1)}} \\ &= q^{-k\binom{d}{2}} \prod_{i=1}^d \prod_{j=1}^k \frac{1 - q^{-(n+i+j-1)}}{1 - q^{-(i+j-1)}}. \end{aligned}$$

Therefore,

$$\text{RHS (2.1)} = q^{kd(n+d-1)-2k\binom{d}{2}} \prod_{i=1}^d \prod_{j=1}^k \frac{1 - q^{-(n+i+j-1)}}{1 - q^{-(i+j-1)}} = \prod_{i=1}^d \prod_{j=1}^k \frac{1 - q^{n+i+j-1}}{1 - q^{i+j-1}}.$$

Finally, we know from the definition (1.1) that

$$\text{LHS (2.1)} = \left\langle \begin{matrix} n+k \\ k \end{matrix} \right\rangle_q^{(d)} = \prod_{j=0}^{d-1} \frac{\begin{bmatrix} n+k+j \\ k \end{bmatrix}_q}{\begin{bmatrix} k+j \\ k \end{bmatrix}_q} = \prod_{j=1}^d \prod_{i=1}^k \frac{1 - q^{n+i+j-1}}{1 - q^{i+j-1}},$$

thereby giving the claimed relation (2.1) by interchanging the indices i and j . $\square$

In view of (2.1), we have

$$\sum_{n \geq 0} \left\langle \begin{matrix} n+k \\ k \end{matrix} \right\rangle_q^{(d)} x^n = \sum_{T \in \text{SYT}(k^d)} q^{\text{comaj}(T)-k\binom{d}{2}} \sum_{n \geq 0} \begin{bmatrix} (n+d-1) - \text{des}(T) + kd \\ kd \end{bmatrix}_q x^n.$$

To simplify the inner sum on the right-hand side, we need to determine the range of $\text{des}(T)$.

Lemma 2.2. *For standard Young tableaux $T \in \text{SYT}(k^d)$, we have*

$$d-1 \leq \text{des}(T) \leq k(d-1). \quad (2.3)$$

Moreover, both inequalities are attainable, each by a unique standard Young tableau.

Proof. For the lower bound, we assume that in the standard Young tableau T , the entries in the first column are $u_1 < u_2 < \dots < u_d$. For each $u_{i+1} - 1$ with $1 \leq i \leq d-1$, it is clear that its row index must be smaller than that of u_{i+1} , thereby implying that $u_{i+1} - 1$ is a descent. Hence,

$$\text{des}(T) \geq |\{u_2 - 1, u_3 - 1, \dots, u_d - 1\}| = d-1.$$

On the other hand, suppose the entries in the last column are $u'_1 < u'_2 < \dots < u'_d$. It is also true that all $u'_1, \dots, u'_{d-1}$ are descents. To ensure $\text{des}(T) = d-1$, we must have $u'_i = u_{i+1} - 1$ for $1 \leq i \leq d-1$. In this situation, the only possibility is the

standard Young tableau $T^\dagger$ of shape (k^d) in which the entry in the i -th row and j -th column is $j + (i - 1)k$.

For the upper bound, we assume that the first row of the standard Young tableau T is filled with $v_1 < v_2 < \dots < v_k$. For each $v_{j+1} - 1$ with $1 \leq j \leq k - 1$, its row index is at least 1, which is never smaller than that of v_{j+1} . Thus,

$$\text{des}(T) \leq |\{1, 2, \dots, kd - 1\} \setminus \{v_2 - 1, v_3 - 1, \dots, v_k - 1\}| = k(d - 1).$$

Suppose further that the entries in the last row are $v'_1 < v'_2 < \dots < v'_k$. It is clear that $v'_1, \dots, v'_{k-1}$ are non-descents. Thus, to ensure $\text{des}(T) = k(d - 1)$, we must have $v'_j = v_{j+1} - 1$. Then the only possible standard Young tableau $T^\ddagger$ is such that the (i, j) -th box is filled with $i + (j - 1)d$. $\square$

According to (2.3), for every $T \in \text{SYT}(k^d)$,

$$(d - 1) - \text{des}(T) \leq 0.$$

Thus,

$$\sum_{n \geq 0} \left[\begin{matrix} (n + d - 1) - \text{des}(T) + kd \\ kd \end{matrix} \right]_q x^n = x^{\text{des}(T) - (d - 1)} \sum_{i \geq 0} \left[\begin{matrix} i + kd \\ kd \end{matrix} \right]_q x^i.$$

Now we can apply the following evaluation of q -binomial series [1, p. 36, eq. (3.3.7)]:

$$\sum_{i \geq 0} \left[\begin{matrix} i + j \\ j \end{matrix} \right]_q x^i = \frac{1}{(x; q)_{j+1}},$$

from which we get

$$\sum_{n \geq 0} \left[\begin{matrix} (n + d - 1) - \text{des}(T) + kd \\ kd \end{matrix} \right]_q x^n = \frac{x^{\text{des}(T) - (d - 1)}}{(x; q)_{kd+1}}.$$

Finally, we conclude that

$$\sum_{n \geq 0} \left\langle \begin{matrix} n + k \\ k \end{matrix} \right\rangle_q^{(d)} x^n = \frac{1}{(x; q)_{kd+1}} \sum_{T \in \text{SYT}(k^d)} q^{\text{comaj}(T) - k \binom{d}{2}} x^{\text{des}(T) - (d - 1)},$$

thereby producing the desired relation (1.3). In addition, we have the coefficient of the lowest power of x :

$$c_{k,0}^{(d)}(q) = q^{\text{comaj}(T^\dagger) - k \binom{d}{2}} = 1,$$

while the coefficient of the highest power of x is

$$c_{k,(k-1)(d-1)}^{(d)}(q) = q^{\text{comaj}(T^\ddagger) - k \binom{d}{2}} = q^{2 \binom{k}{2} \binom{d}{2}};$$

here the standard Young tableaux $T^\dagger$ and $T^\ddagger$ are as in the proof of Lemma 2.2.

3. MAIN THEOREM, PART II: PALINDROMICITY

A direct indication of the relation (1.3) and the bounds (2.3) is that Cigler's series $C_k^{(d)}(x, q)$ is in $\mathbb{N}[q, q^{-1}][x]$, while the highest exponent of x is $(k - 1)(d - 1)$. Now we shall show that the coefficients $c_{k,j}^{(d)}(q)$ of x^j in $C_k^{(d)}(x, q)$ are indeed *polynomials* in $\mathbb{N}[q]$ instead of Laurent polynomials. For this purpose, we have to bound $\text{comaj}(T)$ for standard Young tableaux $T \in \text{SYT}(k^d)$.

Lemma 3.1. *For standard Young tableaux $T \in \text{SYT}(k^d)$, we have*

$$\text{comaj}(T) \geq k \binom{d}{2}. \quad (3.1)$$

Proof. We rewrite $\text{comaj}(T)$ as

$$\text{comaj}(T) = \sum_{i \in \text{Des}(T)} (kd - i) = \sum_{m=1}^{kd} |\{l \in \text{Des}(T) : l < m\}|.$$

Now we prove that, for each m , there are at least $r_m - 1$ descents smaller than m , where m is placed in the (r_m, s_m) -th box. In the s_m -th column, we assume that the first r_m entries are $u_1 < u_2 < \dots < u_{r_m} =: m$. It is clear that in each interval $\{u_i, u_i + 1, \dots, u_{i+1} - 1\}$ with $1 \leq i \leq r_m - 1$, there exists at least one descent l_i . Moreover, $l_i \leq u_{i+1} - 1 \leq u_{r_m} - 1 < m$. Since there are $r_m - 1$ such nonintersecting intervals, we have

$$|\{l \in \text{Des}(T) : l < m\}| \geq r_m - 1.$$

Finally,

$$\text{comaj}(T) \geq \sum_{m=1}^{kd} (r_m - 1) = k \sum_{i=1}^d (i - 1) = k \binom{d}{2},$$

as claimed. $\square$

Now we have seen that

$$C_k^{(d)}(x, q) \in \mathbb{N}[q][x].$$

Moreover, the coefficients $c_{k,j}^{(d)}(q)$ are given by

$$c_{k,j}^{(d)}(q) = \sum_{\substack{T \in \text{SYT}(k^d) \\ \text{des}(T) = j + d - 1}} q^{\text{comaj}(T) - k \binom{d}{2}}.$$

To show these coefficients are palindromic, we need the following observation.

Lemma 3.2. *Define a map $\theta : \text{SYT}(k^d) \rightarrow \text{SYT}(k^d)$ by*

$$\theta(T)(r, s) := kd + 1 - T(d + 1 - r, k + 1 - s).$$

Then θ is an involution. In addition,

$$\text{des}(\theta(T)) = \text{des}(T), \quad (3.2)$$

and

$$\text{comaj}(\theta(T)) = kd \text{des}(T) - \text{comaj}(T). \quad (3.3)$$

Proof. It is a direct verification that θ is an involution on $\text{SYT}(k^d)$. Now we check the two relations on statistics. First,

$$\text{Des}(\theta(T)) = \{kd - i : i \in \text{Des}(T)\},$$

so that (3.2) holds. Next,

$$\text{comaj}(\theta(T)) = \sum_{i' \in \text{Des}(\theta(T))} (kd - i') = \sum_{i \in \text{Des}(T)} i,$$

thereby leading us to (3.3). $\square$

Note that the map θ , if restricted to the subset of standard Young tableaux

$$\{T \in \text{SYT}(k^d) : \text{des}(T) = j + d - 1\},$$

is still an involution. Moreover,

$$\left(\text{comaj}(T) - k \binom{d}{2}\right) + \left(\text{comaj}(\theta(T)) - k \binom{d}{2}\right) = kd(j + d - 1) - 2k \binom{d}{2} = kdj,$$

which is independent of T . These facts tell us that in $c_{k,j}^{(d)}(q)$, the sum of the lowest and highest exponents of q is $k dj$. Also, by pairing T and $\theta(T)$, we have

$$c_{k,j}^{(d)}(q) = q^{kdj} c_{k,j}^{(d)}(q^{-1}),$$

and hence conclude the palindromicity.

4. CLOSING REMARKS

In [2], Cigler also considered a family of Fibonacci–Hoggatt numbers. For nonzero numbers a and b , let $(F_{a,b}(n))_{n \geq 0}$ be a sequence defined by the recurrence

$$F_{a,b}(n) = aF_{a,b}(n-1) + bF_{a,b}(n-2),$$

together with the initial values $F_{a,b}(0) = 0$ and $F_{a,b}(1) = 1$. When $a = b = 1$, the sequence reduces to the conventional *Fibonacci numbers*.

In analogy to (1.1), define the *Fibonacci–Hoggatt numbers* by

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle_{F_{a,b}}^{(d)} := \frac{\langle n \rangle_{F_{a,b}}^{(d)}!}{\langle k \rangle_{F_{a,b}}^{(d)}! \langle n-k \rangle_{F_{a,b}}^{(d)}!} = \prod_{j=0}^{d-1} \frac{\left[\begin{matrix} n+j \\ k \end{matrix} \right]_{F_{a,b}}}{\left[\begin{matrix} k+j \\ k \end{matrix} \right]_{F_{a,b}}},$$

where

$$\langle n \rangle_{F_{a,b}}^{(d)} := \left[\begin{matrix} n+d-1 \\ d \end{matrix} \right]_{F_{a,b}}, \quad \langle n \rangle_{F_{a,b}}^{(d)}! := \prod_{j=1}^n \langle j \rangle_{F_{a,b}}^{(d)},$$

with

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{F_{a,b}} := \begin{cases} \prod_{j=0}^{k-1} \frac{F_{a,b}(n-j)}{F_{a,b}(k-j)}, & \text{if } 0 \leq k \leq n, \\ 0, & \text{otherwise.} \end{cases}$$

Note that the numbers $F_{a,b}(n)$ can be expressed by *Binet's formula*:

$$F_{a,b}(n) = \frac{\alpha^n - \beta^n}{\alpha - \beta},$$

where

$$\alpha := \frac{a + \sqrt{a^2 + 4b}}{2}, \quad \beta := \frac{a - \sqrt{a^2 + 4b}}{2}.$$

Putting

$$q := \frac{\beta}{\alpha},$$

and noting that $\alpha\beta = -b$, we have

$$F_{a,b}(n) = \left(-\frac{b}{q}\right)^{\frac{n-1}{2}} \frac{1 - q^n}{1 - q},$$

so that

$$\begin{bmatrix} n \\ k \end{bmatrix}_{F_{a,b}} = \left(-\frac{q}{b}\right)^{\frac{k(k-n)}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

In [2], Cigler considered the following series

$$G_{k,F_{a,b}}^{(d)}(x) := \left(\sum_{m=0}^{kd+1} (-1)^{\binom{m+1}{2}} b^{\binom{m}{2}} \begin{bmatrix} kd+1 \\ m \end{bmatrix}_{F_{a,b}} x^m \right) \left(\sum_{n \geq 0} \left\langle \begin{matrix} n+k \\ k \end{matrix} \right\rangle_{F_{a,b}}^{(d)} x^n \right). \quad (4.1)$$

If we put

$$y := \left(-\frac{b}{q}\right)^{\frac{kd}{2}} x,$$

it follows from a direct computation that

$$G_{k,F_{a,b}}^{(d)}(x) = (y; q)_{kd+1} \sum_{n \geq 0} \left\langle \begin{matrix} n+k \\ k \end{matrix} \right\rangle_q^{(d)} y^n,$$

where we have applied the following evaluation [1, p. 36, eq. (3.3.6)] to the first sum in (4.1):

$$\sum_{i=0}^j \begin{bmatrix} j \\ i \end{bmatrix}_q (-1)^i z^i q^{\binom{i}{2}} = (z; q)_j.$$

Note that y and x only differ by a scalar. By virtue of Theorem 1.1, we may immediately confirm the following conjecture of Cigler [2, Conjecture 18], which also reduces to [2, Conjecture 14] at the $a = b = 1$ specialization.

Theorem 4.1. $G_{k,F_{a,b}}^{(d)}(x)$ is a polynomial in x of degree $(k-1)(d-1)$ with constant term being 1 and the coefficient of the highest power of x being $b^{2\binom{k}{2}\binom{d}{2}}$.

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REFERENCES

1. G. E. Andrews, *The theory of partitions*, Cambridge University Press, Cambridge, 1998. [5](#), [8](#)
2. J. Cigler, Pascal triangle, Hoggatt matrices, and analogous constructions, preprint, 2021. Available at arXiv:2103.01652. [1](#), [2](#), [7](#), [8](#)
3. D. C. Fielder and C. O. Alford, On a conjecture by Hoggatt with extensions to Hoggatt sums and Hoggatt triangles, *Fibonacci Quart.* **27** (1989), no. 2, 160–168. [1](#)
4. G. Gasper and M. Rahman, *Basic hypergeometric series. Second edition*, Cambridge University Press, Cambridge, 2004. [1](#)
5. R. P. Stanley, A chromatic-like polynomial for ordered sets, in: *Proc. Second Chapel Hill Conf. on Combinatorial Mathematics and its Applications (Univ. North Carolina, Chapel Hill, N.C., 1970)*, 421–427, University of North Carolina, Chapel Hill, NC, 1970. [2](#)
6. R. P. Stanley, Ordered structures and partitions, Thesis (Ph.D.) Harvard University. 1971. [2](#)
7. R. P. Stanley, *Ordered structures and partitions*, American Mathematical Society, Providence, RI, 1972. [2](#)
8. R. P. Stanley, *Enumerative combinatorics. Vol. 2. Second edition*, Cambridge University Press, Cambridge, 2024. [4](#)

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